

Chapter 1

The Derivative

Definition 1.1 (Differentiability). A function f is differentiable at a point x if the following limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For this limit to exist, both the left and right derivatives must exist and be equal.

There are alternative, equivalent ways to express the derivative limit. For instance, letting $y = x + h$, we can write:

$$f'(x) = \lim_{y \rightarrow x} \frac{f(y) - f(x)}{y - x}.$$

Additionally, if a function is already known to be differentiable at x , we can evaluate the derivative using the symmetric difference quotient:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h}.$$

Remark. The existence of the symmetric limit does *not* imply differentiability. For example, if $f(x) = |x|$, the symmetric limit at $x = 0$ evaluates to 0, even though the standard derivative does not exist.

One can continue in this way and define *higher order derivatives*, $f^{(2)}, f^{(3)}, \dots$ and so on, either indefinitely (in which case we say that the function is *infinitely differentiable* and is in C^∞) or up to some finite order, say, $n \in \mathbb{N}$.

Example 1.2. We define, for $n \geq 0$, the functions $f_n : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f_n(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

We will analyse when f_n is differentiable, and for how many times. In the case $n = 0$, one checks that f_0 is not continuous at zero (observe that the function oscillates infinitely many times on any interval $(-\delta, \delta)$ about zero, so intuitively the function never settles on a number as $x \rightarrow 0$).

For $n = 1$, though continuous at zero, the function f_1 is not differentiable at 0 because when we evaluate the quotient, for $x \neq 0$, we have

$$\frac{f_1(x) - f_1(0)}{x - 0} = \frac{x \sin\left(\frac{1}{x}\right)}{x} = \sin\left(\frac{1}{x}\right),$$

which, as we just mentioned, does not have a limit as $x \rightarrow 0$.

For $n \geq 2$, the function is differentiable. Given $n \geq 2$, the quotient evaluates to

$$\frac{f_n(x) - f_n(0)}{x - 0} = x^{n-1} \sin\left(\frac{1}{x}\right),$$

and since $n - 1 > 0$, $x^{n-1} \rightarrow 0$ so using the fact that $|\sin x| \leq 1$ for all x , the squeeze theorem yields

$$\lim_{x \rightarrow 0} \frac{f_n(x) - f_n(0)}{x - 0} = 0$$

for $n \geq 2$.

As an exercise, one can in fact prove that f_n is differentiable exactly $n - 1$ times for $n \geq 2$.

Example 1.3. Let $f(x) = x^2$. We compute the derivative at x using the standard limit definition:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \lim_{h \rightarrow 0} (2x + h) = 2x. \end{aligned}$$

Differentiability is stronger than continuity:

Theorem 1.4. *If $f(x)$ is differentiable at $x = x_0$, then it is also continuous there.*

Proof. Suppose f is differentiable at x_0 . We want to show that $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, which is equivalent to showing $\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = 0$. We manipulate the limit:

$$\lim_{x \rightarrow x_0} (f(x) - f(x_0)) = \lim_{x \rightarrow x_0} \left(\frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \right).$$

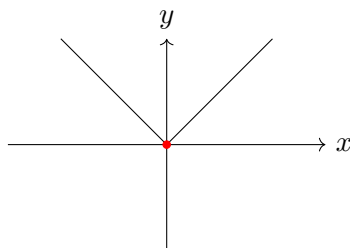
Using the algebraic properties of limits, this separates into:

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot \lim_{x \rightarrow x_0} (x - x_0) = f'(x_0) \cdot 0 = 0.$$

Thus, $f(x) \rightarrow f(x_0)$ as $x \rightarrow x_0$, meaning f is continuous at x_0 . $\square$

Remark. The converse of this theorem is not true; a continuous function is not necessarily differentiable.

Example 1.5. Consider the function $f(x) = |x|$ at $x = 0$.



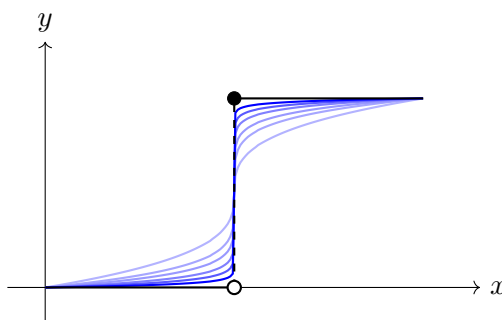
The function is continuous at $x = 0$ because $\lim_{x \rightarrow 0} |x| = 0 = f(0)$. However, evaluating the left and right derivatives yields:

$$\lim_{h \rightarrow 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1,$$

$$\lim_{h \rightarrow 0^-} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1.$$

Since the left and right limits are not equal, the two-sided limit does not exist, and $f(x)$ is not differentiable at $x = 0$.

Consider functions with jump discontinuities. Attempting to approximate them with smooth, polynomial functions provides useful intuition for their derivatives. If we were to connect a discontinuous jump with an approximating continuous curve, the curve would require an incredibly steep slope downwards (or upwards) at the discontinuity. As the approximation becomes exact, the connecting slope becomes vertical, meaning the right (or left) slope does not exist and tends to $\pm\infty$. A slope of $\pm\infty$ represents a vertical tangent line. This idea is illustrated below.



Example 1.6. Suppose the position of a particle moving along a straight line at time t (in seconds) is given by the function $s(t) = t^2 + 3t$ (in meters).

The average speed of the particle over the time interval $[1, 1 + \Delta t]$ is the average rate of change of position:

$$v_{\text{avg}} = \frac{s(1 + \Delta t) - s(1)}{\Delta t} = \frac{((1 + \Delta t)^2 + 3(1 + \Delta t)) - (1^2 + 3(1))}{\Delta t}.$$

Expanding the numerator yields:

$$v_{\text{avg}} = \frac{1 + 2\Delta t + (\Delta t)^2 + 3 + 3\Delta t - 4}{\Delta t} = \frac{5\Delta t + (\Delta t)^2}{\Delta t} = 5 + \Delta t \quad (\text{for } \Delta t \neq 0).$$

This tells us the average speed over any small interval starting at $t = 1$. To find the *instantaneous* speed of the particle at exactly $t = 1$, we take the limit as the interval length Δt approaches 0:

$$v(1) = s'(1) = \lim_{\Delta t \rightarrow 0} (5 + \Delta t) = 5 \text{ m/s}.$$

Example 1.7. Geometrically, the derivative represents the slope of the tangent line to a curve at a given point. Consider the curve defined by $f(x) = x^3 - 2x$. We wish to find the equation of the tangent line to this curve at $x = 2$.

First, we find the coordinates of the point of tangency by evaluating the function:

$$f(2) = (2)^3 - 2(2) = 8 - 4 = 4.$$

So, the line passes through the point $(2, 4)$. Next, we calculate the derivative $f'(x)$ to find the formula for the slope at any point x :

$$f'(x) = \frac{d}{dx}(x^3 - 2x) = 3x^2 - 2.$$

Evaluating the derivative at $x = 2$ gives the slope m of the tangent line:

$$m = f'(2) = 3(2)^2 - 2 = 12 - 2 = 10.$$

Using the point-slope form of a linear equation, $y - y_1 = m(x - x_1)$, the equation of the tangent line is:

$$y - 4 = 10(x - 2) \implies y = 10x - 16.$$

1.1 Properties of the Derivative

Proposition 1.8. Let f, g be differentiable functions. Then

1. (Linearity): $f + g$ is differentiable with $(f + g)' = f' + g'$.

2. (Linearity): cf is differentiable for all $c \in \mathbb{R}$ with $(cf)' = cf'$.
3. (Product Rule): fg is differentiable with $(fg)' = f'g + fg'$.
4. (Quotient Rule): $\frac{f}{g}$ is differentiable, provided $g(x) \neq 0$ for all x , with $\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$.

Example 1.9. As a consequence of the above results, polynomials are differentiable, with

$$\frac{d}{dx} \sum_{k=0}^n a_k x^k = \sum_{k=1}^n k a_k x^{k-1}.$$

Theorem 1.10 (Chain Rule). *Let f, g be differentiable such that g is defined on the range of f . Then $g \circ f$ is differentiable with*

$$\frac{d}{dx} g(f(x)) = g'(f(x)) f'(x)$$

for all x .

Example 1.11. Let $f(x) = \sin(\sqrt{1+x^2})$. Using the chain rule twice (first on $\sin(g(x))$ where $g(x) = \sqrt{1+x^2}$, and then on $g(x) = \sqrt{h(x)}$ where $h(x) = 1+x^2$), we get

$$f'(x) = \frac{d}{dx} \sin(\sqrt{1+x^2}) = \cos(\sqrt{1+x^2}) (\sqrt{1+x^2})' = \frac{x \cos(\sqrt{1+x^2})}{\sqrt{1+x^2}}.$$

1.2 Implicit Differentiation

A useful consequence of the chain rule is the concept of implicit differentiation.

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